

Review the article

Linguistic Interpretability of Transformer-based Language Models

1. Introduction to the Literature Review

The rapid advancement of transformer-based language models has fundamentally reshaped the field of Natural Language Processing (NLP). Models such as BERT, GPT, and their successors have achieved remarkable performance across a wide variety of tasks, including syntactic parsing, semantic analysis, question answering, and text generation. Despite these successes, the internal mechanisms by which these models process and represent linguistic information remain largely opaque. This lack of transparency has motivated a growing body of research focused on linguistic interpretability, which aims to understand how linguistic knowledge is encoded, structured, and utilized within transformer-based architectures. This literature review examines five influential and representative works in this area, including survey papers and recent research published from 2022 onward. The selected studies span probing-based analysis, representation-level analysis, and mechanistic or causal interpretability. For each work, key contributions are summarized, followed by a comparative analysis of methodologies and a critical discussion of strengths and limitations.

2. Rai et al. (2024): A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models

Rai et al. (2024) provide a comprehensive and practice-oriented survey of mechanistic interpretability (MI) for transformer-based language models. Unlike representation-based approaches that focus on correlational patterns, this work frames interpretability as the task of reverse-engineering the internal computations of neural networks. The authors define MI as the identification of the causal components and computational circuits responsible for specific model behaviors. The paper systematically categorizes existing MI research into several themes, including feature discovery, circuit analysis, activation patching, and causal interventions. It emphasizes that understanding a model requires not only identifying where information is stored, but also determining how different components interact to produce behavior. The survey highlights successful case studies where small transformer models exhibit interpretable circuits for tasks such as numerical comparison or pattern matching. A major strength of this work is its clear articulation of open challenges, particularly scalability to large language models (LLMs) and the dependence of results on carefully designed tasks. However, the paper also reveals a key limitation of current MI approaches: many findings are highly model- and task-specific, making generalization difficult. Nevertheless, this survey serves as an essential roadmap for future research seeking causal explanations of linguistic behavior in transformers.

3. Belinkov (2022): Probing Classifiers: Promises, Shortcomings, and Advances

Belinkov (2022) offers a critical examination of probing classifiers, one of the most widely used tools for analyzing linguistic representations in neural models. Probing involves training a lightweight classifier on frozen model representations to predict linguistic properties such as part-of-speech tags, syntactic dependencies, or semantic roles. High probing accuracy is often interpreted as evidence that the model encodes the corresponding linguistic information. The central contribution of this work lies in clarifying the limitations of probing-based interpretations. Belinkov argues that probing primarily measures extractability, not usage. In other words, the fact that a linguistic feature can be extracted from a representation does not imply that the model uses that feature during prediction. The paper discusses methodological pitfalls such as overly powerful probes, lack of control tasks, and misinterpretation of results. The strength of this work lies in its methodological rigor and its role in establishing best practices for probing studies. However, probing remains inherently correlational and cannot establish causal relationships. As such, while probing is useful for mapping where linguistic information may reside, it must be complemented by causal methods for stronger interpretability claims.

4. Rogers et al. (2020): A Primer in BERTology: What We Know About How BERT Works

Rogers et al. (2020) present a seminal survey of research analyzing BERT, often referred to as “BERTology.” This paper synthesizes a large body of work investigating what linguistic knowledge BERT encodes, how that knowledge is distributed across layers, and how it can be accessed through probing and representation analysis. One of the most influential ideas discussed in this survey is the layerwise linguistic hierarchy hypothesis, which suggests that lower layers of BERT encode surface and syntactic information, while higher layers capture more abstract semantic and discourse-level features. The paper also reviews evidence for BERT’s ability to represent syntactic structure, lexical semantics, and world knowledge. The primary strength of this survey is its breadth and integrative perspective, which makes it a foundational reference for linguistic interpretability research. However, its

focus on BERT-era models limits its applicability to newer instruction-tuned and generative LLMs. Nonetheless, many conceptual insights from BERTology continue to inform contemporary interpretability research.

5. Conmy et al. (2023): Towards Automated Circuit Discovery for Mechanistic Interpretability

Conmy et al. (2023) represent a significant step toward operationalizing mechanistic interpretability through automated circuit discovery. The authors introduce a framework called ACDC, which aims to identify minimal causal subgraphs—or circuits—within transformer models that are responsible for specific behaviors. The proposed approach leverages activation patching to systematically test the causal contribution of model components. By iteratively removing or retaining components based on their causal impact, the method identifies compact and interpretable circuits. The authors demonstrate their approach on tasks such as numerical comparison in GPT-2 Small, successfully recovering known computational structures. The key strength of this work is its movement toward scalability and automation in mechanistic interpretability. However, the approach remains sensitive to task design and evaluation metrics, and its applicability to complex, real-world linguistic behaviors is still an open question. Despite these limitations, the paper marks a crucial advance toward systematic causal analysis of transformer models.

6. Starace et al. (2023): Probing LLMs for Joint Encoding of Linguistic Categories

Starace et al. (2023) extend traditional probing approaches by investigating whether multiple related linguistic categories are jointly encoded within shared model representations. Rather than probing single features in isolation, the authors examine interactions between syntactic categories, such as part-of-speech classes and corresponding dependency relations. Their results provide evidence that certain linguistic phenomena are encoded in a structured and interdependent manner, suggesting a richer internal organization of linguistic knowledge than previously assumed. The study also reports similar patterns across multilingual language models, indicating partial cross-lingual generalization. The primary contribution of this work is its move beyond single-feature probing toward a more holistic analysis of linguistic structure. However, like other probing-based studies, it remains correlational and does not establish whether jointly encoded features play a causal role in downstream predictions.

7. Comparison of Methods

Across the reviewed literature, two dominant interpretability paradigms emerge. Probing-based and representation-level approaches treat interpretability as an information-retrieval problem, asking whether linguistic properties can be extracted from model representations. These methods are scalable and well-suited for broad exploratory analysis but are limited in their ability to support causal claims. In contrast, mechanistic and causal interpretability approaches conceptualize interpretability as reverse-engineering computation. By intervening on internal components and analyzing causal effects, these methods provide stronger explanatory power. However, they face challenges related to scalability, task dependence, and computational cost, particularly for large language models.

8. Strengths and Weaknesses of the Literature

Overall, the literature demonstrates substantial progress in understanding how transformers encode linguistic information. Strengths include methodological diversification, increased awareness of causal limitations, and deeper integration of linguistic theory. At the same time, several weaknesses remain, including the lack of standardized benchmarks, limited generalization of findings, and the high cost of causal analysis.

9. Summary

In summary, research on linguistic interpretability has matured from exploratory probing studies to more rigorous causal and mechanistic analyses. While no single approach provides a complete solution, the combination of probing, representation analysis, and mechanistic interpretability offers a promising path toward transparent and trustworthy language models. Future work must focus on scalability, generalization, and integration with large language models to fully realize the goals of linguistic interpretability.

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